

Linear Regression:

ML Pipeline: Get Class $\mathcal{Y}$, Training loss $\mathcal{L}$, Optim. Method

Multiple Lin Reg: loss fcts: $\mathcal{L}_{\text{sq}}(f(x), y) = (y - f(x))^2$ if $|f(x) - y| \leq \delta$, $\mathcal{L}_{\text{hinge}}(f(x), y) = \delta(1 - |f(x) - y| - \delta)$ if $|f(x) - y| > \delta$, $\mathcal{L}_{\text{p}}(f(x), y) = \tau \max\{0, y - f(x)\} + (1 - \tau) \max\{0, f(x) - y\}$, $\tau \in [0, 1]$

$\hookrightarrow \tau \in [0, \frac{1}{2}] \Rightarrow f$ underestimates, $\tau \in (\frac{1}{2}, 1] \Rightarrow f$ overestimates

LinReg Estimators: $\hat{w} = \arg\min_w \|y - Xw\|_2^2 = (X^T X)^{-1} X^T y$

$\hat{y} = X\hat{w} = X(X^T X)^{-1} X^T y = \Pi_X y$ ("projection onto span(X)")

$\hookrightarrow d > n \Rightarrow X^T X$ not invertible $\Rightarrow \hat{w}$ not unique bec. $\dim(\text{Ker}(X^T X)) > 0$

$\hookrightarrow$ Min Norm Solution: $\hat{w} = \min_w \|w\|_2$ s.t. $y = Xw$

$\Rightarrow \hat{w} = X^T(XX^T)^{-1}y = (X^T X)^{-1}X^T y$ here $(X^T X)^{-1} = X(X^T X)^{-1}X^T$

Nonlinear LS: $\hat{w} = \arg\min_w \|y - \Phi w\|_2^2 = (\Phi^T \Phi)^{-1} \Phi^T y$

$\Phi^T \Phi$ invertible $\Leftrightarrow \text{rank}(\Phi) = p$, $\Phi^T \Phi(w) = \Phi^T \Phi \Rightarrow$ posdef. $\Rightarrow$ unique

Remarks: $\hat{w} = (\hat{w}_1, \dots, \hat{w}_p)^T$ & $\hat{r} = (r_1, \dots, r_n)^T \Rightarrow w_0 = \hat{w}$, $w_1 = \hat{r}$

Optimization: Iterative Approach: $\odot$ Comp cheap if $\Phi^T \Phi$ is small, $\ominus$ might not conv.

GD: $w^{t+1} = w^t - \eta \nabla \mathcal{L}(w)$

Conv. of GD for LinReg: GD converges for $\min_w \|y - Xw\|_2^2$

if $\eta \leq \frac{2}{\lambda_{\max}(X^T X)}$, $\eta_{\text{opt}} = \frac{2}{\lambda_{\min} + \lambda_{\max}}$ well conditioned if $\lambda_{\min} \approx \lambda_{\max}$

Other Grad-Based Methods: 2nd Order: use Hessian of $\mathcal{L}$

Momentum: $w^{t+1} = w^t + \eta \nabla \mathcal{L}(w^t) - \eta \nabla \mathcal{L}(w^{t-1})$

Adaptive: $w_i^{t+1} = w_i^t - \eta \frac{\partial \mathcal{L}}{\partial w_i}(w^t)$, $\hat{S}_i^t = (w_i^t - w_i^{t-1})^2$

Stochastic GD: random subset 1 datapoint k use for GD

(Mini)Batch SGD: $S \subseteq \{1, \dots, n\}$, $\bar{\nabla} \mathcal{L}_S(w) = \frac{1}{|S|} \sum_{i \in S} \nabla \mathcal{L}(f_i(x), y_i)$

$\hookrightarrow$ In practice: $D \subseteq S$ u. $U \subseteq S$ k train Model on S 's

$\hookrightarrow$ $|S|$ larger $\Rightarrow$ Var $\downarrow$ & Comp. Compl. $\uparrow$

Remarks: $w^* = (X^T X)^{-1} X^T y$ if $\Phi = X$, $\hat{w} = (X^T X)^{-1} X^T y$

$\hookrightarrow A = V(\Lambda - \gamma I)^{-1} V^T$ (Ray Mat to Span(X)), $B = V(\frac{\gamma}{2})^{-1} U^T y$

Convexity: $f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y) \Leftrightarrow \nabla^2 f(x)$ posd. $\forall x$

$\Leftrightarrow f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle \Leftrightarrow$

Lemma: (i) f, g conv, $\alpha, \beta > 0 \Rightarrow \alpha f + \beta g$ conv, (ii) max{f, g} conv, if ∇f convex & ∇g affine or ∇f convex & ∇g affine

(iii) f, g convex: if $\{f \text{ (convex)} \Rightarrow \text{non-decreasing}, g \text{ convex}\}$

(iv) f convex, diff., x^* stat point $\Rightarrow x^*$ global min (if f strongly convex)

Strongly Convex with $\mu > 0 \Leftrightarrow f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y) - \frac{\mu}{2} \lambda(1-\lambda) \|x - y\|^2$

Strongly Convex $\Leftrightarrow$ strictly convex, $\nabla^2 f = \mu$ posd. $\Rightarrow \lambda(\nabla^2 f) \geq \mu$

Model Evaluation & Selection:

Estim Err: $\mathcal{L}(\hat{f}(x), f(x))$, Pred. Err: $\mathcal{L}(\hat{f}(x), y)$

Gener. Err.: Average Pred Error: $\frac{1}{n} \sum_{i=1}^n \mathcal{L}(\hat{f}(x_i), y_i)$

Estim of GE: Empirical Risk: $\mathcal{L}(f; D) = \frac{1}{n} \sum_{i=1}^n \mathcal{L}(f(x_i), y_i)$

$\Rightarrow \hat{f}_D = \arg\min_{f \in \mathcal{F}} \mathcal{L}(f; D)$, GE too opt. naive: if train based on D

$D = D_{\text{train}} \cup D_{\text{test}}$, $|D_{\text{train}}| = |D_{\text{test}}| = \frac{n}{2}$, $|D_{\text{train}}| > |D_{\text{test}}|$ better for model training, but test error is not very accurate, $|D_{\text{train}}| = |D_{\text{test}}|$ worse for training

but better estim of GE. (bec. choice $\hat{f} \neq \hat{f}_D$ $\Rightarrow$ GE $\neq$ $\hat{f}_D$)

Cross Validation: If $D = D_{\text{train}} \cup D_{\text{test}}$ $\Rightarrow \hat{f}$ still depends on D_{test}

Model Seld. w val set: $D = D_{\text{train}} \cup D_{\text{val}} \cup D_{\text{test}}$ $\Rightarrow$ Cannot give estim of GE

$\hookrightarrow D_{\text{train}}$ for training, use D_{val} to check for least error among $\hat{f}$

D_{test} to estim GE

k-fold CV: $D_{\text{train}} = D_{\text{val}} \cup \dots \cup D_{\text{test}}$. Use D_{val} for train

k calc. val loss on D_{val} . Repeat $\forall$ D_{val} & D_{test} $\Rightarrow$ obtain avg. $\mathcal{L}(w_k)$

$\mathcal{L}_k = \mathcal{L}(\hat{f}_k, D_{\text{test}}) = \frac{1}{|D_{\text{test}}|} \sum_{(x,y) \in D_{\text{test}}} \mathcal{L}(\hat{f}_k(x), y)$

$\Rightarrow$ Crossvalidation Error: $CV_k(\mathcal{M}) = \frac{1}{k} \sum_{i=1}^k \mathcal{L}_i$

$\Rightarrow$ Select $\mathcal{M}_k$ with least $CV_k(\mathcal{M}_k) \Rightarrow$ train $\mathcal{M}_k$ on whole D_{train}

$\hat{f} = \hat{f}_{\mathcal{M}_k, D_{\text{train}}}$ & estim GE as $\mathcal{L}(\hat{f}; D_{\text{test}}) = \frac{1}{|D_{\text{test}}|} \sum_{(x,y) \in D_{\text{test}}} \mathcal{L}(\hat{f}(x), y)$

LeaveOneOut CV: $K = |D_{\text{test}}|$, $|D_{\text{train}}| = 1 \Rightarrow$

K_i Submodels closer to full model BUT overprec. of $\mathcal{L}_k \approx 6E$

Bias-Variance Tradeoff & Regular.

$\mathcal{F}$ too simple $\Rightarrow$ underfitting

$\mathcal{F}$ too complex $\Rightarrow$ overfitting

CV-Tradeoff: $\hat{f}_D = \mathcal{M}(D)$

Model Bias: $\text{Bias}_D(\hat{f}_D, x) = (E_D[\hat{f}_D(x)] - f(x))^2$

$\text{Bias}_D(\hat{f}_D) - E_X[\text{Bias}_D(\hat{f}_D, x)] =$ Simple Model: Large Bias
Complex Model: Small Bias

Model Variance: $\text{Var}_D(\hat{f}_D, x) = E_D[(\hat{f}_D(x) - E_D[\hat{f}_D(x)])^2]$

$= E_D[(\hat{f}_D(x))^2] - (E_D[\hat{f}_D(x)])^2 = \text{Var}_D(\hat{f}_D) = E_X[\text{Var}_D(\hat{f}_D, x)]$

Approx: $E_D[\hat{f}_D] \approx \bar{f} = \frac{1}{n} \sum_{i=1}^n f_i$, $\text{Var}_D(\hat{f}_D) \approx \frac{1}{n} \sum_{i=1}^n (f_i - \bar{f})^2$

$\hookrightarrow$ Simple Model: Small Var., Complex Model: Large Var.

CV-Decomp: $E_D[\mathcal{L}(\hat{f}_D; R_{\mathcal{F}})] = \text{Var}_D(\hat{f}_D) + \text{Bias}_D(\hat{f}_D)^2$

$E_D[(\hat{f}_D(x) - f(x))^2] = \text{Var}_D(\hat{f}_D, x) + \text{Bias}_D(\hat{f}_D, x)^2$

Lasso Regr.: $\min_w \|y - Xw\|_2^2$ s.t. $\|w\|_1 \leq R$ (Small R $\Rightarrow$ Large λ)

$\Leftrightarrow \min_w (\|y - Xw\|_2^2 + \lambda \|w\|_1) \Rightarrow$ If $X^T X = \mathbf{I}$ $\Rightarrow \hat{w}_\lambda = \text{sign}(w_0) \max\{|w_0| - \lambda, 0\}$

$\Rightarrow$ Lasso induces Sparsity ($\beta_j = 0$). If input dim's highly correlated

(columns of X) $\Rightarrow$ Lasso might not be stable

Ridge Regr.: $\min_w \|y - Xw\|_2^2$ s.t. $\|w\|_2 \leq R$ $\Rightarrow$ $\min_w (\|y - Xw\|_2^2 + \lambda \|w\|_2^2)$

$\hookrightarrow w_{\lambda, \text{ridge}} = (X^T X + \lambda I)^{-1} X^T y = \frac{1}{1 + \lambda} w^0$, $\text{Bias}_D(w_{\lambda, \text{ridge}}) = \lambda \|\hat{f}_D\|_2^2$

$\text{Bias}_D(\hat{f}_D) - \|w_{\lambda, \text{ridge}}\|_2^2 = \frac{1}{1 + \lambda} \|w^0\|_2^2$, $\text{Var}_D(\hat{f}_D) = \sigma^2 \sum_{i=1}^n \frac{1}{1 + \lambda}$

$\Rightarrow 1 - 0 \cdot \left\{ \text{Var} = \frac{\sigma^2}{1 + \lambda} \right\}$, $\lambda \rightarrow \infty \Rightarrow \text{Bias} \rightarrow 0$, $\lambda \rightarrow 0 \Rightarrow \text{Var} \rightarrow \infty$

Choice of λ : GE reads min for a $\hat{\lambda}$: find $\hat{\lambda}$ with CV

Remark: $\|w\|_2 \leq \|w\|_1 \leq \sqrt{2} \|w\|_2$

Classification: labels are discrete, binary, CL:

Binary CL: $f: \mathbb{R} \rightarrow \{-1, 1\}$, if $f: \mathbb{R} \rightarrow \mathbb{R} \Rightarrow \hat{y} = \text{sign}(f(x))$

0-1 loss: $\mathcal{L}_{0-1}(y, \hat{y}) = \mathbb{1}_{y \neq \hat{y}} = \frac{1}{2} (1 - y\hat{y}) = \frac{1}{2} (1 - y_0 y_1)$

$\hookrightarrow$ Decision Boundary: $\{x \in \mathbb{R}^d : \hat{f}(x) = 0\}$

Featurized Lin Classif.: $\Phi: \mathbb{R}^d \rightarrow \mathbb{R}^p$, $\hat{f}_w = \text{sign}(w^T \Phi(x))$

$\hookrightarrow$ Decision Boundary: $\{x \in \mathbb{R}^d : w^T \Phi(x) = 0\}$

Surrogate Loss fct for training:

Surrogate $\mathcal{L}$ for train. & loss for val.: for non convex & hard to

used for $\hat{f}_D$ 1 Accuracy: $\frac{1}{|D_{\text{test}}|} \sum_{(x,y) \in D_{\text{test}}} \mathbb{1}_{\hat{f}_D(x) = y}$. Surrogate loss should be:

convex, differentiable, only depend on $z = y \cdot f(w, x)$, decreasing

λ bound from below ($\mathcal{L}(z) \geq \mathcal{L}_{\text{0-1}}(z)$, $\mathcal{L}_{\text{0-1}}(z) = \log(1 + e^{-z})$

$\frac{\partial}{\partial z} \mathcal{L}_{\text{0-1}}$ is unbounded but $\frac{\partial}{\partial z} \mathcal{L}_{\text{0-1}} \in (-1, 1) \Rightarrow \mathcal{L}_{\text{0-1}}$ better

Logistic Regression: $\mathcal{L}(w) = \frac{1}{n} \sum_{i=1}^n \log(1 + e^{-y_i w^T x_i})$

Max-Margin Solution & Logistic Regression:

$\mathcal{L}_{\text{0-1}}$ is convex & diff. $\Rightarrow$ possible to find minimizer $\hat{w}$ if (!)

the data is linearly separable. $\Rightarrow \mathcal{L} \rightarrow 0$ but $\hat{w}$ not unique

Max Margin Sol: $\hat{w}_{\text{max}} = \arg\max_{w} \text{margin}(w) = \arg\max_{w} \min_{(x,y) \in D} y \cdot w^T x$

Hard SVM: $\hat{w}_{\text{max}} = (\min_{(x,y) \in D} \|w\|_2 \text{ s.t. } y \cdot w^T x \geq 1, y = \pm 1, \dots)$

Remarks: (i) $y \cdot (w^T x) \geq |K(x, x)|$, $\|w\|_2 = \|K(x, x)\|_2 \cos(\theta)$

(ii) For linearly separable D : $w_{\text{min}} = \frac{1}{\|w_{\text{max}}\|_2}$

Satisfies $\lim_{t \rightarrow \infty} \frac{w_t}{\|w_t\|_2} = \hat{w}_{\text{max}}$ margin $(\hat{w}) = \frac{1}{\|\hat{w}\|_2}$

Soft-Margin Sol: D not linearly sep. ($\exists i: y_i \cdot w^T x_i < 0$)

& $\ker(X) \neq \emptyset \Rightarrow \exists$ global minimizer $\hat{w}$ of min hinge (w)

GD with $\eta = \frac{4}{\lambda_{\max}(X)}$ satisfies $\lim_{t \rightarrow \infty} w^t = \hat{w}$

Def $\hat{\mathcal{S}} = (S_1, \dots, S_n) \in \mathbb{R}^n \Rightarrow$ Require: $y_i \cdot \langle \hat{w}, \hat{\mathcal{S}}_i \rangle \geq 1 - S_i$

Optim Problem: $\min_{w \in \mathbb{R}^d} (\|w\|_2^2 + \lambda \sum_{i=1}^n S_i)$ s.t. $y_i \cdot \langle w, \hat{\mathcal{S}}_i \rangle \geq 1 - S_i$

$= \min_{w \in \mathbb{R}^d} [\|w\|_2^2 + \lambda \sum_{i=1}^n \max(0, 1 - y_i \cdot \langle w, \hat{\mathcal{S}}_i \rangle)]$

Remarks: $\# \{S_i > 0\} = \#$ wrongly classif pts & $\# \{x: x^T w \in [1, 2]\}$

$S_i = 0 \forall$ correctly classifd pts outside $x^T w = \pm 1$ margin

Multiclass Classif.: Train K diff. linear Models

$\hat{f}_1, \hat{f}_2 \in \mathbb{R}_K$. $\forall \hat{f}_i \Rightarrow \hat{y} = \arg\max_{k \in K} \hat{f}_k(x)$

1vsRest: Train K binary Classifiers: $\hat{y} = \begin{cases} 1 & y=1 \\ k & y=k \end{cases}$

Cross-Entropy: fit $\hat{f}_1, \dots, \hat{f}_K$ simultaneously $(\hat{f}_1(x), \dots, \hat{f}_K(x))$

$\mathcal{L}_{\text{CE}}(\hat{f}(x), y) = -\log(\frac{1}{\sum_{k=1}^K \hat{f}_k(x)}) = \hat{f} = \arg\min_{\hat{f}} \sum_{i=1}^n \mathcal{L}_{\text{CE}}(\hat{f}(x_i), y_i)$

In practice: Kernelized ALS: $\min_{w \in \mathbb{R}^d} \|y - Xw\|_2^2 \Rightarrow \min_{w \in \mathbb{R}^d} \|y - Xw\|_2^2 \Rightarrow \min_{w \in \mathbb{R}^d} \|y - Xw\|_2^2$

Kern Ridge Reg: $\min_w \|y - Xw\|_2^2 + \lambda \|w\|_2^2 = \min_w \|y - Xw\|_2^2 + \lambda \|w\|_2^2$

$\hat{w} = \arg\min_w [\sum_{i=1}^n \mathcal{L}_{\text{CE}}(f_i(x_i), y_i)]$

Approx of GE: $\text{GE} = \frac{1}{|D_{\text{test}}|} \sum_{(x,y) \in D_{\text{test}}} \mathcal{L}_{\text{0-1}}(y, \hat{f}(x))$

Other evaluation Metrics for Classifiers:

Assign the "important" class to -1 & other CL +1

$H_0: \hat{y}(w) = -1$, $H_1: \hat{y}(w) = +1$

If FN & FP not equally severe: $\text{GE} = \frac{\text{FN}}{\text{FN} + \text{FP}} + \frac{\text{FP}}{\text{FN} + \text{FP}}$

Minimizing this cost only depends on $S = \frac{\text{FN}}{\text{FP}}$ $\Rightarrow$ Usually $S \rightarrow 1 \Rightarrow \frac{\text{FN}}{\text{FP}} \rightarrow 1$

$\text{FNR} = \frac{\text{FN}}{\# \{y = +1\}} = P(\hat{y}(w) = -1 | y = +1)$, $\text{FOR} = \frac{\text{FP}}{\# \{y = -1\}} = P(\hat{y}(w) = +1 | y = -1)$

$\text{FPR} = \frac{\text{FP}}{\# \{y = -1\}} = P(\hat{y}(w) = +1 | y = -1)$, $\text{Accuracy} = \frac{\text{TP} + \text{TN}}{n}$

$\text{TPR} = \text{Sensitivity} = \text{power} = \text{recall} = \frac{\text{TP}}{\# \{y = +1\}}$ 1-FOR

$\text{TNR} = \text{Specificity} = \frac{\text{TN}}{\# \{y = -1\}}$, $\text{Precision} = \frac{\text{TP}}{\# \{y = +1\}}$

$\text{F1-Score} = \frac{2}{\frac{1}{\text{recall}} + \frac{1}{\text{precision}}}$ want large, want small

$\hat{y}_2(x) = \text{sign}(\hat{f}_2(x)) = \begin{cases} +1 & \text{if } \hat{f}_2(x) > 0 \\ -1 & \text{if } \hat{f}_2(x) \leq 0 \end{cases}$

ROC Curve: Receiver Operating Characteristic

Area Under the ROC Curve (AUROC): measure of Classifier Perform. $\in [0, 1]$

$\hat{y}_2(x) = \text{sign}(\hat{f}_2(x)) = \begin{cases} +1 & \text{if } \hat{f}_2(x) > 0 \\ -1 & \text{if } \hat{f}_2(x) \leq 0 \end{cases}$

Distribution Shifts & Robustness: Priority group: max $\{y_1 z_1, y_2 z_2, \dots\}$

Fairness: Equal Odds: $\forall y, s \in \{1, 2\} P(y=1 | x=1, s) = P(y=1 | x=2, s)$

Demographic Parity: $\forall s \in \{1, 2\} P(y=1 | s) = P(y=1 | s)$

Equal Opportunity: $P(y=1 | s=1, y=1) = P(y=1 | s=2, y=1)$

Remark: Equal Op. & Dem Parity: accuracy-fairness-tradeoff

Kernels: Kernelization: Assume $\hat{w} = \Phi^T \tilde{w} \in \mathbb{R}^p$, $\tilde{w} \in \mathbb{R}^p$

$\hat{w} \approx \tilde{w} = \min_{\tilde{w} \in \mathbb{R}^p} \sum_{i=1}^n \mathcal{L}(y_i, \sum_{j=1}^p \tilde{w}_j \Phi(x_i, x_j)) = \min_{\tilde{w} \in \mathbb{R}^p} \sum_{i=1}^n \mathcal{L}(y_i, \langle \tilde{w}, \Phi(x_i) \rangle)$

Replace $\langle \Phi(x), \Phi(y) \rangle = K(x, y)$, $\Phi \in \mathbb{R}^{n \times p}$, $K \in \mathbb{R}^{n \times n}$

Valid Kernel: (i) Symmetric: $K(x, x) = K(x, x)$, (ii) K posd.

Thm: 1) $\Phi: \mathbb{R}^d \rightarrow \mathbb{R}^p$, $\tau: \mathbb{R}^p \rightarrow \mathbb{R}^p \Rightarrow K(x, x) = \langle \tau(\Phi(x)), \tau(\Phi(x)) \rangle$ valid ker.

2) k_1, k_2 valid $\Rightarrow K(x, x) = k_1(x, x) + k_2(x, x)$ & $K(x, y) = k_1(x, y) + k_2(x, y)$ valid

3) k_1, k_2 valid $\Rightarrow K(x, x) = k_1(x, x) \cdot k_2(x, x)$ & $K(x, y) = k_1(x, y) \cdot k_2(x, y)$ valid

Validity Check: 1) Symmetry, 2) valid comb., 3) easily not posd., 4) feature map

Examples: g Taylor series with coeff 20: $K(x, y) = g(x, y)$ valid

Polynomial Kernel: $K(x, y) = (1 + x \cdot y)^m \Rightarrow \Phi(x) \in \mathbb{R}^p$, $p = \binom{m+d-1}{d-1}$

Gaussian Kernel: $K(x, y) = \exp(-\frac{\|x - y\|_2^2}{2\sigma^2})$ $\hookrightarrow$ bandwidth, $\sigma = 2 \log \frac{1}{\epsilon}$

Thm: $\mathcal{L}$ strongly convex & smooth, $\hat{w} = \arg\min_{w \in \mathbb{R}^d} \sum_{i=1}^n \mathcal{L}(w, x_i) \in \mathbb{R}^d$

$\Rightarrow \exists \tilde{w} \in \mathbb{R}^d: \hat{w} = \tilde{w}$, $\tilde{w}$ global minimizer of kernelized obj. fct.

Kernelized ALS: $\min_{w \in \mathbb{R}^d} \|y - Xw\|_2^2 \Rightarrow \min_{w \in \mathbb{R}^d} \|y - Xw\|_2^2 \Rightarrow \min_{w \in \mathbb{R}^d} \|y - Xw\|_2^2$

Classification: $\mathcal{L}(y_i, \hat{f}(x_i)) = \log(1 + e^{-y_i \hat{f}(x_i)}) \Rightarrow \hat{w} = \arg\min_w \frac{1}{n} \sum_{i=1}^n \log(1 + e^{-y_i \hat{f}(x_i)})$

Remarks: False: $\forall k \exists \hat{y}: k \neq \hat{y}(x_1, x_2)$, $\hat{y}(x) = \text{sign}(\sum_{j=1}^n \hat{y}_j K(x, x_j))$

$k = \frac{\max}{\min}$ not valid, k can be negative, $k = \min \forall x, y > 0$ valid, $k = \max$ not valid

$k = \frac{\min}{\max} \forall x, y > 0$ valid, $k = \cos(x - y)$ valid, $k = (x \cdot y)^2$ not valid

$k = |A \cap B|$ valid, $k = 2|A \cap B| - |A| - |B|$ not valid

Neural Networks Idea: learn Φ as well $\rightarrow \Phi(x) = \phi(\phi(x))$

with ϕ a non linear activation fct, e.g. $\phi(x) = \tanh(x)$, $\phi_{\text{sig}}(x) = \frac{1}{1 + e^{-x}}$, $\phi_{\text{relu}}(x) = \max(0, x)$

Thm: $\mathcal{S} \rightarrow \mathbb{R}$ cont. fct & ϕ any sigmoidal fct ($\phi(x) \rightarrow 0$, $\phi(x) \rightarrow 1$, $\phi(x) \rightarrow \infty$). Then

$\hat{f}$ can be uniformly approx by $\hat{f}(x) = \sum_{i=1}^n w_i \phi(w_i^T x + b_i)$

Backward propagation: Optim Problem is non-convex! $\Rightarrow$ Gradient Descent

Optimization in a general NN:

Setup: $W = [W^{(1)}, \dots, W^{(L)}]$ $\Rightarrow$ Goal: $\hat{w}_L = (w_L^1, \dots, w_L^m)$

Def: $\delta = 2(f(x; W) - y)$, $z^{(l)} = W^{(l)} v^{(l-1)}$, $v^{(l)} = \phi(z^{(l)})$

1) $(\hat{w}_{l+1})^T = \frac{\partial \mathcal{L}}{\partial w^{(l+1)}} = \frac{\partial \mathcal{L}}{\partial w^{(l+1)}} \frac{\partial \mathcal{L}}{\partial w^{(l+1)}}$ (Note: $\frac{\partial \mathcal{L}}{\partial w^{(l+1)}} = \frac{\partial \mathcal{L}}{\partial w^{(l+1)}}$)

Note: $\frac{\partial \mathcal{L}}{\partial w} = \frac{\partial \mathcal{L}}{\partial w} = \frac{\partial \mathcal{L}}{\partial w} \in \mathbb{R}^n \Rightarrow \frac{\partial \mathcal{L}}{\partial w} = \frac{\partial \mathcal{L}}{\partial w}$

$\hookrightarrow f = W^{(l)} v^{(l-1)} \Rightarrow \frac{\partial \mathcal{L}}{\partial w^{(l)}} = \frac{\partial \mathcal{L}}{\partial w^{(l)}} = \frac{\partial \mathcal{L}}{\partial w^{(l)}}$ Note: $\delta^{(l-1)}$ calculated in forward pass

$\hookrightarrow (\hat{w}_{l+1})^T = \frac{\partial \mathcal{L}}{\partial w^{(l+1)}} = \delta^{(l+1)}$

2) $(\hat{w}_{l+1})^T = \frac{\partial \mathcal{L}}{\partial w^{(l+1)}} = \frac{\partial \mathcal{L}}{\partial w^{(l+1)}} \frac{\partial \mathcal{L}}{\partial w^{(l+1)}}$ $\delta = W^{(l)} \frac{\partial \mathcal{L}}{\partial w^{(l-1)}}$

$\frac{\partial \mathcal{L}}{\partial w^{(l-1)}} = \begin{bmatrix} \frac{\partial \mathcal{L}}{\partial w^{(l-1)}} & \frac{\partial \mathcal{L}}{\partial w^{(l-1)}} & \dots & \frac{\partial \mathcal{L}}{\partial w^{(l-1)}} \end{bmatrix} \in \mathbb{R}^n$

3) $(\hat{w}_{l+1})^T = \frac{\partial \mathcal{L}}{\partial w^{(l+1)}} = \frac{\partial \mathcal{L}}{\partial w^{(l+1)}} \frac{\partial \mathcal{L}}{\partial w^{(l+1)}}$ Note: $\frac{\partial \mathcal{L}}{\partial w^{(l-1)}} = \frac{\partial \mathcal{L}}{\partial w^{(l-1)}}$

$\Rightarrow (\hat{w}_{l+1})^T = \frac{\partial \mathcal{L}}{\partial w^{(l+1)}} = \frac{\partial \mathcal{L}}{\partial w^{(l+1)}} \frac{\partial \mathcal{L}}{\partial w^{(l+1)}}$

Optimization & training techniques: Convergence depends on initial.

SGD better than GD bec. it can escape local min.

Vanishing & Exploding Grad: Optim fails if $\|w\|_2 \rightarrow 0 \Rightarrow$ control $\phi'(z)$

$\phi(z) = \frac{1}{1 + e^{-z}} \Rightarrow \phi'(z) = \phi(z)(1 - \phi(z))$ $\Rightarrow$ only good if $z \approx 0$, $\text{ReLU}(x) = \max(0, x) \Rightarrow \phi'(x) = \begin{cases} 1 & x > 0 \\ 0 & x \leq 0 \end{cases}$

Random Weight Init: For 1-layer NN ideal: $w \sim \mathcal{N}(0, \frac{2}{n})$ "dot-input dim"

Heuristics: 1) Xavier init: $w \sim \mathcal{N}(0, \frac{2}{n_1 + n_2})$ or $w \sim \mathcal{N}(0, \frac{2}{n_1})$, 2) He init: $w \sim \mathcal{N}(0, \frac{2}{n_1})$

Learning Rate & weight optim: Ideally: $\eta \rightarrow 0$, better: $\frac{1}{\eta} \propto \frac{\|w\|_2}{\|w\|_2}$

Regularization in NNs: 1) Penalty term: $\arg\min_w \sum_{i=1}^n \mathcal{L}(w, x_i) + \lambda \|w\|_2^2$

2) Early Stopping: e.g. Stop if val loss increases

3) Dropout: Randomly omit units with prob. $1-p$ in each iteration of SGD

$\hookrightarrow$ Train remaining subnetwork $\Rightarrow$ For testing use all units, adjust $\hat{w}_i \cdot p_i \rightarrow \hat{w}_i$

4) Batch Normalization: Correct for mean drift $\rightarrow$ Standardize batches

Algo: 1) $\hat{w} = \frac{1}{n} \sum_{i=1}^n w_i$ for 1 unit, 2) learnable params: β, γ , 3) $y_i = \hat{y}_i - \beta + \gamma \hat{y}_i$

$\mu_0 = \frac{1}{n} \sum_{i=1}^n \mu_i$, $\sigma^2 = \frac{1}{n} \sum_{i=1}^n (\mu_i - \mu_0)^2 \Rightarrow \hat{y}_i = \frac{\mu_i - \mu_0}{\sqrt{\sigma^2}}$, $c > 0 \Rightarrow \hat{y}_i = \gamma \hat{y}_i + \beta$

4) Output v_j, v_{j+1}

Convolutional Neural Networks

Images can be represented as $n \times m \times c$ tensor of color
Idea: Units share some weights, & each unit connected to few units in the next layer

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$$

1D Conv: $w \in \mathbb{R}^h, x \in \mathbb{R}^d \rightarrow w \otimes x \in \mathbb{R}^{h \times d}$

CNN: ANN with $\nabla^{(i)} = (\varphi^{(i)} + \psi^{(i)})$ layers, but may also contain fully connected layers

Remark: Weights are called filters.

Multi-dimensional Convolution:

Consider $n \times m \times 3$ RGB image with m filters of size $f \times g$.

Padding: Add a frame (couple of white row/cols) to the image $\Rightarrow$ new dim: $(n+2p) \times (m+2p)$

Stride: Stepsize of filter. Must match dim of filter: $(n+2p) \% s = 0$

Output Dim: $\left(\frac{n+2p}{s} + 1\right) \times \left(\frac{m+2p}{s} + 1\right) \times m$

Clustering (Unsupervised Learning)

Unsupervised learning: learning without y 's.

Goal: extract meaningful patterns. (Clustering $\leftrightarrow$ Classification)

Standard Clustering: group D into clusters based on similarity

Hierarchical Clust: 1) Form Individual Clusters $\forall D$ points

2) Iteratively merge pairs of clusters that are closest

k-Means Clustering & Lloyd's Heuristic

Each Cluster represented by its center $\bar{z}_i \in \mathbb{R}^d$

Def: Cluster assignment: $z_i = \arg \min_j \|x_i - \bar{z}_j\|_2$

Goal: learn $\bar{z}_j$ in training $\Rightarrow$ assign z_i in testing based on $\bar{z}_j$

$$\bar{z} = \frac{1}{n} \sum_{i=1}^n x_i$$

Lloyd's Heuristic: Iteratively update $z_i, \bar{z}_j$

Def: $\bar{z}^{(t)} = (\bar{z}_1^{(t)}, \dots, \bar{z}_K^{(t)})$

1) Init $\bar{z}^{(0)}$, 2) Repeat: $\bar{z}_i^{(t+1)} = \arg \min_j \sum_{x \in C_j} \|x - \bar{z}_i^{(t)}\|_2$

Convergence Analysis & Limitations:

Thm: $\bar{R}(\bar{z}^{(t+1)}) \leq \bar{R}(\bar{z}^{(t)})$

Lloyd's Algorithm guaranteed to conv.

But only to local min. Θ : Selecting 'K' often problem

Can extend to Supervised case

Init Schemes & k-Means++

Farthest point: $x_i \sim \mathcal{U}(D) \Rightarrow \mu_i = \arg \min_{x \in D} \|x - \mu_i\|_2$

k-Means++ 1) $\mu_i = x_i \in D$, 2) $\mu_j = \arg \min_{x \in D} \|x - \mu_i\|_2$

Chasing K: Increasing k always yields better results!

Heuristics: Elbow Heur., Adding Regul.: $\bar{R}(\bar{z}) + \lambda \|\bar{z}\|_2$

PCA: Goal: $x_1, \dots, x_n \in \mathbb{R}^d \rightarrow z_1, \dots, z_n \in \mathbb{R}^k, k \ll n$

k=1: wgt $\bar{z}_1 \propto v$. Goal: $\forall x \in \mathbb{R}^d$ find $z \in \mathbb{R}^k$ s.t. $\bar{z} \approx x$

Minimize reconstr. err: $\min_{\bar{z}} \sum_{i=1}^n \|x_i - \bar{z}\|_2^2$ s.t. $\|\bar{z}\|_2 = 1$

$$\bar{W}^* = \arg \min_{W \in \mathbb{R}^{d \times k}} \sum_{i=1}^n \|x_i - Wz_i\|_2^2 \Rightarrow W^* = \arg \min_{W \in \mathbb{R}^{d \times k}} \|X - WZ\|_2^2$$

$$L^* = \arg \min_{W \in \mathbb{R}^{d \times k}} \sum_{i=1}^n (w_i^T x_i)^2 = \arg \min_{W \in \mathbb{R}^{d \times k}} W^T X X^T W$$

Remarks: $W^* \in \mathbb{R}^{d \times k}$ is not unique, if $\lambda_1 = \dots = \lambda_k = \lambda_{k+1}$

Any convex comb. of $\bar{z}_i$'s also optm $\Rightarrow$ not unique

1st ksd: $\bar{W}_1, \dots, \bar{W}_k = \arg \min_{W_1, \dots, W_k} \sum_{i=1}^n \|x_i - W_i z_i\|_2^2$

Optimal k: Use CV. Heuristic: k s.t. 35% of var explained

SVD: $X = U S V^T \rightarrow n, \bar{Z} = U V^T \rightarrow$ first k EV of X

PCA & k-Means: Both Matrix factor. Probl.

$$PCA: W^*, \bar{z}_1, \dots, \bar{z}_k = \arg \min_{W \in \mathbb{R}^{d \times k}} \sum_{i=1}^n \|x_i - W \bar{z}_i\|_2^2$$

$$k-M: W^*, \bar{z}_1, \dots, \bar{z}_k = \arg \min_{W \in \mathbb{R}^{d \times k}} \sum_{i=1}^n \|x_i - W \bar{z}_i\|_2^2$$

Kernel PCA: $x \in \mathbb{R}^d \rightarrow \phi(x) \in \mathbb{R}^D \rightarrow z = W \phi(x), W \in \mathbb{R}^{D \times k}$

Autoencoders: Encoder: $x \mapsto z = W^T x$, Decoder: $z \mapsto \hat{x} = W z$

Goal: find subspace minimizing $\sum_{i=1}^n \|x_i - \hat{x}_i\|_2^2$

Idea: learn identity fct: $x \mapsto f(x) = x$

Probabilistic Modeling

Training data D: ML method efficiently find $\hat{f} \in F$

Algorithmic perspective: $\hat{f}(x) = \mathcal{U}(D)$

Probabilistic perspective: $\hat{f}(x) = \mathcal{U}(D)$

Def: $\hat{f}(x) = \mathcal{U}(D)$

Remarks: MAP w/ Gaussian Prior: $\hat{\theta} \sim \mathcal{U}(0, 1)$

Bay. Opt. Cl. using $\log$: $\hat{f}(x) = \mathbb{E}[Y|X=x]$

Remarks: MAP w/ Gaussian: $P_{\bar{z}_j} = \mathcal{N}(\mu_j, \Sigma_j)$

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Histogram

Kernel Density Estimator: $\hat{p}_n(x) = \frac{1}{n} \sum_{i=1}^n \delta(x - x_i)$

Statistical Inference: Paradigms: Frequentist vs. Bayesian

Strong prior belief $\Rightarrow$ θ unobserved rand. var. $\sim P_0$

$$p(\theta|D) = \frac{p(D|\theta)p(\theta)}{p(D)}, p(\theta) = \int p(D|\theta)p(\theta) d\theta$$

Maximum Likelihood Estimation: $\hat{\theta} = \arg \max_{\theta \in \Theta} p(\theta|D)$

Def: KL Divergence: measure of "difference" between ref. dist. P & another dist. Q

$$D_{KL}(P||Q) = \mathbb{E}_{X \sim P} [\log \frac{p(X)}{q(X)}] = \int p(x) \log \frac{p(x)}{q(x)} dx$$

Remarks: $D_{KL}(P||Q) + D_{KL}(Q||P) \geq 0$

Maximum A Posteriori Estimation: $\hat{\theta}_{MAP} = \arg \max_{\theta \in \Theta} p(\theta|D)$

$$\hat{\theta}_{MAP} = \arg \max_{\theta \in \Theta} p(\theta) \prod_{i=1}^n p(x_i|\theta)$$

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Probabilistic Classification: Discriminative & Generative

Discriminative Model: model cond. prob. $Y|X$ ($P_{Y|X}$)

Generative Model: Model $P_{X,Y}$ by $P_{X|Y}$ & P_Y

Classification w/ Discriminative Models: Focus on binary Cl.: $y \in \{1, 2\}$

Model $p(y|x)$ by $\hat{g}(x) = \frac{1}{1 + e^{-f(x)}}$ for $Y|X = x \sim \text{Bern}(p(y|x))$

$$p(y|x) = \sigma(y \cdot f(x)) = \frac{1}{1 + e^{-y \cdot f(x)}}$$

MLE: $\hat{\theta}_{MLE} = \arg \min_{\theta \in \Theta} \sum_{i=1}^n \log(1 + e^{-y_i \cdot f(x_i; \theta)})$

MAP: Gaussian prior: $\hat{\theta}_{MAP} \sim \mathcal{N}(0, \Sigma)$, Laplace prior: $\hat{\theta}_{MAP} \sim \text{Lap}(0, \lambda)$

$$\hat{\theta}_{MAP} = \arg \min_{\theta \in \Theta} \sum_{i=1}^n \log(1 + e^{-y_i \cdot f(x_i; \theta)}) + \frac{\lambda}{2} \|\theta\|_2^2$$

LP: $\hat{\theta}_{MAP} = \arg \min_{\theta \in \Theta} \sum_{i=1}^n \log(1 + e^{-y_i \cdot f(x_i; \theta)}) + \frac{1}{b} \|\theta\|_1$

Classification w/ generative Models: Aim: $p(y|x) = \frac{p(x,y)}{p(x)} \propto p(x|y)p(y) \Rightarrow$ estim. $p(y|x)$

Gaussian Naive Bayes Model: $P(Y=y) = p_y$

'Naive': each dim indep.: $p(x_1, \dots, x_n|y) = \prod_{i=1}^n p(x_i|y)$

$$x_j|Y=y \sim \mathcal{N}(\mu_{j,y}, \sigma_{j,y}^2) \Rightarrow \hat{z}_j = \log(\sigma_{j,y}^2)$$

MLE: $\hat{\mu}_j = \frac{1}{n} \sum_{i=1}^n x_{ij}, \hat{\sigma}_{j,y}^2 = \frac{1}{n} \sum_{i=1}^n (x_{ij} - \hat{\mu}_j)^2$

Binary Cl.: $\hat{y} = \text{sign}(\log \frac{p(1|x)}{p(2|x)}) = \log \frac{p(1|x)}{p(2|x)}$

Fischer's LDA: (F)LOA: $\hat{z}_1, \dots, \hat{z}_k, (R_1, \dots, R_k)$

Special Case of GMD: $\hat{e}_1, \dots, \hat{e}_k = \hat{e}_j^* \Rightarrow \hat{y} = \text{sign}(\hat{e}_j^* x)$

$$f(x) = \log \frac{p(1|x)}{p(2|x)} = \sum_{j=1}^k \hat{e}_j^T x$$

Bayes Opt. Cl.: $\hat{f}(x) = \arg \min_{f \in F} \mathbb{E}[L(f, Y)|X=x]$

$$0-1 \text{ Loss: } L(f(x), y) = \begin{cases} 0 & \text{if } f(x) = y \\ 1 & \text{if } f(x) \neq y \end{cases} \Rightarrow \mathbb{E}[L] = 1 - p(x|y)$$

Asym. Cl.: $L(f(x), y) = C_{10} 1_{f(x)=1} + C_{01} 1_{f(x)=0}$

$$\mathbb{E}[L(f(x), y)|X=x] = (1 - p(x|1))C_{10} + p(x|1)C_{01}$$

Binary Cl.: $L(f(x), y) = \begin{cases} 0 & \text{if } f(x) = y \\ 1 & \text{if } f(x) \neq y \end{cases}$

$$\mathbb{E}[L(f(x), y)|X=x] = (1 - p(x|1))C_{10} + p(x|1)C_{01}$$

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Clustering w/ GMMs

EM: $\hat{z}_i = \arg \max_{j \in \{1, \dots, K\}} \frac{w_j \exp(-\frac{1}{2\sigma_j^2} \|x_i - \mu_j\|_2^2)}{\sum_{j=1}^K w_j \exp(-\frac{1}{2\sigma_j^2} \|x_i - \mu_j\|_2^2)}$

Hard EM: E-Step: $\hat{z}_i^{(t)} = \arg \max_{j \in \{1, \dots, K\}} \frac{w_j \exp(-\frac{1}{2\sigma_j^2} \|x_i - \mu_j\|_2^2)}{\sum_{j=1}^K w_j \exp(-\frac{1}{2\sigma_j^2} \|x_i - \mu_j\|_2^2)}$

M-Step: $\hat{\mu}_j^{(t+1)} = \frac{\sum_{i=1}^n \hat{z}_i^{(t)} x_i}{\sum_{i=1}^n \hat{z}_i^{(t)}}, \hat{\sigma}_j^{(t+1)} = \sqrt{\frac{\sum_{i=1}^n \hat{z}_i^{(t)} \|x_i - \hat{\mu}_j^{(t+1)}\|_2^2}{n_j}}$

Soft EM: E-Step: $\hat{z}_i^{(t)} = \frac{w_j \exp(-\frac{1}{2\sigma_j^2} \|x_i - \mu_j\|_2^2)}{\sum_{j=1}^K w_j \exp(-\frac{1}{2\sigma_j^2} \|x_i - \mu_j\|_2^2)}$

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